

Hallucination Failure Types

Project: Verified Study Buddy

1. Hallucination Failure Types

Teach the learning cycle

Primary target: Explain the mechanism and terminology of generation failures and verification

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Multimodal Evidence

Without opening the old lesson, explain one key idea from Multimodal Evidence. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of speech pipelines and multimodal evidence

Reference example, not the only acceptable wording: Teacher explanation: A multimodal pipeline may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly. Track source modality, segment/time references, transcription, interpretation and final claims separately. Compare conflicting evidence without assuming one modality always wins. Ask for clarification when the task needs unresolved information.

Visual Composition

Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of image generation and provenance

Reference example, not the only acceptable wording: Teacher explanation: An image generator constructs an output from learned patterns and a specification. A generated picture is not a record that the depicted event occurred. Keep a provenance record containing specification, tool/version when known, generation date, edits, permissions and intended use. Label fictional imagery. Inspect visual requirements and avoid inferring identity or event authenticity from appearance.

Structured Output and Schemas

Without opening the old lesson, explain one key idea from Structured Output and Schemas. Give an example and a mistake that the idea helps you avoid.

Look for: Separate JSON parsing, schema validation and source support checks

Reference example, not the only acceptable wording: Teacher explanation: Parsing establishes grammatical JSON; schema checks enforce fields/types/conditional rules; semantic checks match the claims to sources and domain meaning. None alone establishes all three.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A generated answer contains a nonexistent quote, a real source unrelated to the claim and a correct sum using the wrong input. Classify the three failures and their repairs.

Concept to explain

Week 14: Hallucination Failure Types

Term · Meaning

Term: unsupported claim

Meaning: A statement for which the supplied evidence does not establish the asserted conclusion.

Term: fabrication

Meaning: Invented information presented as if it were established fact.

Term: faulty inference

Meaning: A conclusion that does not follow from the available premises or evidence.

Term: citation error

Meaning: A reference that is wrong, missing or does not support the claim attached to it.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Two 45-minute sessions. Generated answers may invent facts, misread quantities or cite irrelevant passages. Verification checks each claim against evidence rather than accepting fluent wording. Use the supplied constructed scenarios to make the mechanism inspectable. Collect W4 before sharing solutions. Students should name intermediate evidence and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the inputs, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from predictions or unmeasured conditions. If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

End goal for the pictorial case

By the end, I can distinguish supported, contradicted and unsupported generated claims.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

1. Claim 1 is directly supported by the passage.
2. Claim 2 invents a scientist's quotation: fabrication.
3. Claim 3 draws a cause from correlation: faulty inference.
4. Tag each claim, repair only with evidence, or mark unknown.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

An answer invents a quotation, cites a source lacking it and supplies an impossible date.

Separate the failures and checks.

Exact explanation to model

Check exact quote text, claim-to-source support and calendar consistency independently. Each failure needs its own correction.

Explain the added worked example

Generated answers may invent facts, misread quantities or cite irrelevant passages. Verification checks each claim against evidence rather than accepting fluent wording.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible

explanation as a measured outcome.

Split an answer into atomic claims. For each, record the exact source passage, support status and any needed calculation. Mark unsupported and contradicted claims separately; do not repair them by inventing sources.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term · Meaning

Term: Atomic claim

Meaning: A single factual statement checked separately

Term: Contradicted

Meaning: Inconsistent with the supplied evidence

Term: Unsupported

Meaning: Not established by the available evidence

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Source R says room B seats 12 and the meeting starts at 15:00. Generated answer says B seats 20, starts at 15:00 and includes free lunch.

Capacity 20 is contradicted; 15:00 is supported; free lunch is unsupported. One of three claims has support. The missing lunch fact is not evidence of no lunch.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Hallucination Failure Types

A fictional Study Buddy drafts a field-trip summary from one approved record.

The facts we will use

Source S: trail length 12 km; planned constant speed 4 km/h, excluding stops; departure 09:00.

Draft: “The walk takes two hours, starts at 09:00 and includes free lunch.” No lunch record is supplied.

Hallucination Failure Types

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute duration

$12 / 4 = 3$ hours, excluding stops.



2. Contradicted claim

Two-hour duration disagrees with the calculation from S.



3. Supported claim

Departure 09:00 is stated in S.



4. Unsupported claim

Free lunch has no source; missing evidence is not a confirmed no-lunch fact.

CHANGE ONE THING

Add a planned 30-minute stop while preserving speed and distance.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Compute duration

What happens and why: $12 \div 4 = 3$ hours, excluding stops.

Step: 2. Contradicted claim

What happens and why: Two-hour duration disagrees with the calculation from S.

Step: 3. Supported claim

What happens and why: Departure 09:00 is stated in S.

Step: 4. Unsupported claim

What happens and why: Free lunch has no source; missing evidence is not a confirmed no-lunch fact.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Different errors need different repairs. Recalculate the duration; keep the supported departure; remove or seek evidence for lunch. These categories are more informative than calling the whole answer wrong. Arithmetic inherits assumptions such as constant speed and no stops, so avoid turning the three-hour walking calculation into a guaranteed arrival time.

What this example does not establish

The fictional plan does not establish actual speed, delays or what eventually occurred on a real trip.

Change one thing in the pictured case

Change: Add a planned 30-minute stop while preserving speed and distance.

Prediction: Planned total elapsed time becomes 3.5 hours.

Reason: Three moving hours plus half an hour stopped; walking-time calculation itself stays three.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: An answer invents a quotation, cites a source lacking it and supplies an impossible date.

Write the question to explain: Separate the failures and checks.

Reveal after predictions: Check exact quote text, claim-to-source support and calendar consistency

independently. Each failure needs its own correction.

Keep for an independent attempt: An answer gives a book title that does not exist and quotes words absent from the cited passage. Identify two failures and a repair.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: Capacity 20 is contradicted; 15:00 is supported; free lunch is unsupported. One of three claims has support. The missing lunch fact is not evidence of no lunch. Qualification: Source agreement and arithmetic validity are separate checks; missing evidence is not proof of the opposite claim. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1: 0–7 minutes

Teacher action: Retrieve the prerequisite; identify the input, procedure and evidence source.

Stage: 7–18 minutes

Teacher action: Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.

Stage: 18–32 minutes

Teacher action: Trace the declared procedure and worked case. Complete W2, recording intermediate steps.

Stage: 32–45 minutes

Teacher action: Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.

Stage: Session 2: 0–8 minutes

Teacher action: Retrieve the worked reasoning and collect the guided response before discussing it.

Stage: 8–20 minutes

Teacher action: Compare a proposed boundary variant with the original case; identify what remains fixed.

Stage: 20–35 minutes

Teacher action: Collect W4 independently. Require the procedure trace and the bounded conclusion.

Stage: 35–45 minutes

Teacher action: Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Completion task: Source R says room B seats 12 and the meeting starts at 15:00. Generated answer says B seats 20, starts at 15:00 and includes free lunch. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Capacity 20 is contradicted; 15:00 is supported; free lunch is unsupported. One of three claims has support. The missing lunch fact is not evidence of no lunch.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Generation failures and verification. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Source agreement and arithmetic validity are separate checks; missing evidence is not proof of the opposite claim.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step

for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Can we conclude there will definitely be no lunch from S alone?

Reasoned teacher answer

No. Lunch is unknown from S, not disproved. Say the source does not supply lunch information.

Changed-case reasoning: Planned total elapsed time becomes 3.5 hours. Three moving hours plus half an hour stopped; walking-time calculation itself stays three.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Color-code every claim in a generated paragraph, classify failures, attach supporting passage lines, and rewrite with honest uncertainty.

What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The worked result establishes every future case: Source agreement and arithmetic validity are separate checks; missing evidence is not proof of the opposite claim.

A missing input may be replaced without recording the change: A replacement changes the evidence or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent evidence: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed

incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Check exact quote text, claim-to-source support and calendar consistency independently. Each failure needs its own correction.

Independent question

An answer gives a book title that does not exist and quotes words absent from the cited passage. Identify two failures and a repair.

Expected independent answer

The title is fabricated and the quotation is unsupported. Verify the source, remove or correct the claims, and state when evidence is missing.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “A citation-looking label makes an invented quote trustworthy.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

The first fabricates a quote: retrieve exact text or remove it. The second misuses a citation: find supporting evidence or retract the claim. The third has a faulty input: verify original values before recalculating. One generic 'check sources' instruction is insufficient.

Explanations for the additional workbook examples

W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Teacher explanation: Generated answers may invent facts, misread quantities or cite irrelevant passages. Verification checks each claim against evidence rather than accepting fluent wording. Split an answer into atomic claims. For each, record the exact source passage, support status and any needed calculation. Mark unsupported and contradicted claims separately; do not repair them by inventing sources.

W2 Trace the worked case

Source R says room B seats 12 and the meeting starts at 15:00. Generated answer says B seats 20, starts at 15:00 and includes free lunch. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Capacity 20 is contradicted; 15:00 is supported; free lunch is unsupported. One of three claims has support. The missing lunch fact is not evidence of no lunch.

W3 Audit the limit

A report omits this qualification: Source agreement and arithmetic validity are separate checks; missing evidence is not proof of the opposite claim. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Source agreement and arithmetic validity are separate checks; missing evidence is not proof of the opposite claim. State the observed result separately from the unverified or out-of-scope claim; preserve the original evidence rather than inventing a missing fact.

W4 Apply a new case independently

Source S: trip is 18 km; vehicle speed is 6 km/h, constant, excluding stops. Answer says 2 hours, 18 km and includes lunch. Audit and recalculate.

Teacher explanation: Travel time is $18/6=3$ hours under the stated assumption, so 2 hours is contradicted. Distance supported; lunch unsupported. Do not infer actual duration with unknown stops.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked input or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure. Require preserved settings, original outputs and data roles. Source agreement and arithmetic validity are separate checks; missing evidence is not proof of the opposite claim. The new case should test this particular mechanism rather than merely restate the original result.

Assess independent transfer

Selected case: Source S: trip is 18 km; vehicle speed is 6 km/h, constant, excluding stops. Answer says 2 hours, 18 km and includes lunch. Audit and recalculate.

Selected case answer

Teacher explanation: Travel time is $18/6=3$ hours under the stated assumption, so 2 hours is contradicted. Distance supported; lunch unsupported. Do not infer actual duration with unknown stops.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Verified Study Buddy

Study Buddy must distinguish source-backed claims, calculations, and unverified generation.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record.
Retrieve this record next week rather than restarting the project.

Support and stretch

Provide one short approved passage; separate instruction, input, output and evidence. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 15: Source Quality. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Travel time is $18/6=3$ hours under the stated assumption, so 2 hours is contradicted. Distance supported; lunch unsupported. Do not infer actual duration with unknown stops. Require intermediate reasoning and the stated scope. Fresh retry: Source says 24 km at 8 km/h excluding stops; answer says 4 hours and lunch included. Expected reasoning: Calculated duration 3 hours contradicts 4; lunch unsupported. Real elapsed time with stops remains unknown. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Next connection: Image generation and provenance. Preserve the current case record and retrieve its evidence boundary before moving on.

Plan the delayed return

Suggested return points in this level: ai-3/15, ai-3/17, ai-3/20. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Audit Study Buddy outputs claim by claim, recording contradiction, missing support and correct calculations separately.